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Alireza Nasirzadeh, PhD

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PROFESSIONAL EXPERIENCE

Research Professor

Feb 2022 – Present

*Assistive and Rehabilitation Robotics Laboratory, School of Mechanical Engineering
Chung-Ang University, South Korea*

Biomechanist

Jan 2020 – Oct 2021

Athlete Evaluation Department, I.R.I. National Paralympic Committee, Iran

Assistant Professor

Sep 2019 – Nov 2021

Faculty of Physical Education and Sport Sciences, Kharazmi University, Iran

Teaching Assistant

Jan 2016 – May 2019

Various Universities, Iran

EDUCATION

PhD in Sports Biomechanics

Sep 2012 – Sep 2016

Kharazmi University, Iran

Thesis: Comparison of Selected Kinematical, Kinetical, and Functional Lower Limb Variables during Walking and Running at Different Speeds among Young Male Athletes Focusing on Limb Dominance

MSc in Physical Education and Sports Sciences

Sep 2009 – May 2011

University of Birjand, Iran

Dissertation: Architectural Characteristics of the Upper Extremity Skeletal Muscles in Young Male Table Tennis Players

BSc in Mathematics

Sep 2003 – May 2009

University of Birjand, Iran

RESEARCH AREAS

Movement Biomechanics · Wearable Robotics & Exosuits · Musculoskeletal Modeling · Dynamic Stability and Balance · Rehabilitation Engineering · Injury Mechanics

PUBLICATIONS

Nasirzadeh, A. et al. (2026). [Biomechanical Effects of Passive Exosuit Assistance on Tibiofemoral Loading and Dynamic Stability during Downhill Walking](#). *Journal of NeuroEngineering and Rehabilitation*.

Park, C., **Nasirzadeh, A.**, Lee, K., Kim, G., & Lee, G. (2026). [Real-Time Estimation of User Adaptation During Hip Exosuit-Assisted Walking Using Wearable Inertial Measurement Unit Data and Long Short-Term Memory Modeling](#). *Biomimetics*, 11(2), 96.

Nasirzadeh, A. et al. (2025). [Harnessing Resistance: The Impact of a Passive Exosuit on Golf Swing](#)

[Biomechanics](#). *Journal of Biomechanics*, 192, 112958.

Choe, E., Moon, J., Ryu, J., Yang, S., **Nasirzadeh, A.**, Kong, S., Choi, Y., & Lee, G. (2025). [Low-Profile, Shoe-Type Ankle-Foot Orthosis with Active Variable Ankle Stiffness via Wire-Fabric Compression Mechanism](#). *Biomimetics*, 10(8), 539.

Kim, Y., Park, S., Yang, S., **Nasirzadeh, A.**, Lee, G. (2024). [The Interplay Between Muscular Activity and Pattern Recognition of Electro-Stimulated Haptic Cues During Normal Walking: A Pilot Study](#). *Bioengineering*, 11, 1248.

Nasirzadeh, A. et al. (2023). [The Effects of a Custom-Designed High-Collar Shoe on Muscular Activity, Dynamic Stability, and Leg Stiffness: A Biomimetic Approach Study](#). *Biomimetics*, 8, 274.

Park, J., Nam, K., Yun, J., Moon, J., Ryu, J., Park, S., **Nasirzadeh, A.** et al. (2023). [Effect of hip abduction assistance on metabolic cost and balance during human walking](#). *Science Robotics*, 8(83), eade0876.

Nasirzadeh, A. et al. (2023). [Influence of circumferential ankle pressure of shoe collar on the kinematics, dynamic stability, electromyography, and plantar pressure during normal walking](#). *PLOS ONE*, 18(2), e0281684.

Nasirzadeh, A. (2020). [The influence of horizontal velocity on inter-lower-limbs local and global asymmetry during walking](#). *Physical Treatments*, 10(2), 69-78.

Nasirzadeh, A., Mohseni, F. (2018). [Local and global symmetry during jogging in young male athletes](#). *Physical Treatments*, 8(1), 17-26.

Nasirzadeh, A. et al. (2017). [Gait symmetry and its evaluation methods: a review](#). *Scientific Journal of Rehabilitation Medicine*, 6(2), 283-97.

Nasirzadeh, A. et al. (2017). [A review of selected factors that affect gait symmetry](#). *Physical Treatments*, 7(1), 3-12.

Nasirzadeh, A. et al. (2016). [The influence of movement velocity on inter-lower-limbs kinematical symmetry in normal walking: a pilot study](#). *Scientific Journal of Rehabilitation Medicine*, 5(4), 159-72.

Nasirzadeh, A. et al. (2015). [Multivariate analysis of 200-m front crawl swimming performance in young male swimmers](#). *Acta of Bioengineering and Biomechanics*, 17(3), 137-43.

Nasirzadeh, A. et al. (2014). [Selected anthropometrical, muscular architecture, and biomechanical variables as predictors of 50-m performance of front crawl swimming in young male swimmers](#). *Science and Sports*, 29(5), e75-e81.

Nasirzadeh, A. et al. (2014). [Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers](#). *Journal of Sports Science and Medicine*, 13(3), 550-556.

UNDER REVIEW OR IN PROGRESS

Nasirzadeh, A. (under review). Shank Attachment Geometry Governs Multi-Joint Force Transmission During Handle-Supported Walking. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*.

Kim, G., Lee, S., **Nasirzadeh, A.** (under review). Rapid Hip-Abduction Exosuit-Induced Torque to Improve Mediolateral Standing Balance Without Stepping. *Nature Biomedical Engineering*.

Yang, J., **Nasirzadeh, A.** (under review). Single-Motor Differential Cable Drivetrain for Bilateral Bidirectional Hip Assistance. *ASME Journal of Mechanisms and Robotics*.

PRESENTATIONS

PODIUM PRESENTATIONS

Mediolateral Balance Recovery during Walking using H-Medi Exosuit (2026). **44th International Society of Biomechanics in Sports Conference (ISBS)**, Loughborough, UK.

Design and Validation of a Passive Exosuit to Enhance Golf Swing Performance (2025). **7th International Conference on Medicine in Novel Technology and Devices**, Hangzhou, China.

[Power Up the Swing: A Biomechanical Study Towards a Golf Exosuit](#) (2024). **42nd International Society of Biomechanics in Sports Conference (ISBS)**, Salzburg, Austria.

[Active Variable Compression Shoes: Design and Evaluation of an External Ankle Supporter to Prevent Ankle Sprain via Active Compression Mechanism](#) (2024). **10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)**, Heidelberg, Germany.

The effect of long-term custom-made foot orthoses on foot structures in children with flexible flat feet (2018). **Eleventh International Conference of Sport Sciences**, Tehran, Iran.

[The effect of gender and anthropometrical parameters on the dynamics of sprint start among young elite sprinters](#) (2012). **XXX International Society of Biomechanics in Sports (ISBS) Conference**, Melbourne, Australia.

POSTER PRESENTATIONS

Reducing Tibiofemoral Joint Force During Downhill Walking Using Various Passive Exosuits: A Pilot Study (2025). **XXX Congress of the International Society of Biomechanics (ISB)**, Stockholm, Sweden.

Passive Collar Stiffness of a Custom-designed High-Collar Shoe (2023). **XXIX Congress of the International Society of Biomechanics (ISB)**, Fukuoka, Japan.

Isokinetic analysis of elite Iranian Paralympic athletes (2021). **Second National Olympic & Paralympic Congress**, Tehran, Iran.

Epidemiology of Sports Injuries and Injury Prevention Strategies in Para Powerlifting (2021). **Asian Paralympic Committee** (online).

The effect of a researcher-made soft insole on selected kinematic and kinetic variables of walking of active girls (2015). **First National Conference on New Findings in Sport Sciences**, Tehran, Iran.

The progress of front crawl swimming speed performance in adolescent swimmers emphasizing biomechanics, muscular architecture, and anthropometric factors (2013). **Second National Conference on Sports Talent Identification**, Tehran, Iran.

Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers (2012). **XXXII FMSI World Congress of Sports Medicine**, Rome, Italy.

Ultrasonographic examination of muscle architecture parameters of the upper arm muscles of table tennis players aged 40-60 years (2011). **Sixth National Conference on Physical Education and Sport Sciences**, Tehran, Iran.

Investigating the indicators of muscular architecture in different parts of the brachial muscles of table tennis players (2011). **First International Sports Medicine Congress**, Isfahan, Iran.

TEACHING EXPERIENCE

Instructor · Kharazmi University

2016-2021

Sports Biomechanics, Research Methods in Biomechanics, Statistics in Sports and Exercise, Anatomy and Kinesiology, Advanced Methods in Biomechanics, Motion Capture principles.

Workshops

Physical Fitness Parameters in Children with Disabilities (2018). **I.R.I. Sports Federation for the Disabled.**

Principles of Motion Analysis and Interpreting Results using Visual3D Software (2018). **Kharazmi University.**

Force Plate: Applications in Sport Sciences and Data Analysis (2017). **Shiraz University.**

Biomechanical Assessment of Athletic Performance (2014). **I.R.I. Sports Federation for the Disabled.**

AWARDS & HONORS

Outstanding Student

2016

Department of Sports Biomechanics, Kharazmi University

Outstanding Presentation Award in Biomechanics

2015

First National Conference on New Findings in Sport Sciences

New Investigator Award

2011

Sixth National Conference on Physical Education and Sport Sciences

SERVICE & OUTREACH

Ad-Hoc Reviewer

Journal of NeuroEngineering and Rehabilitation, Journal of Sports Sciences, Sports Biomechanics, International Journal of Athletic Therapy & Training, BMC Musculoskeletal Disorders, Physical Treatments, PLOS ONE, Scientific Reports, IEEE Transactions on Neural Systems & Rehabilitation Engineering.

MEMBERSHIPS & AFFILIATIONS

International Society of Biomechanics

International Society of Biomechanics in Sports

TECHNICAL SKILLS

- **Biomechanical Analysis:** Gait, kinematics, kinetics, EMG
- **Wearable Robotics:** Exosuits and rehabilitation devices
- **Musculoskeletal Modeling:** AnyBody Modeling System
- **Programming:** MATLAB, Python, LabVIEW
- **Experimental Systems:** Vicon, Qualisys, force plates, IMUs

ENGLISH LANGUAGE SKILLS

PTE Academic Overall 81/90 - Listening 85, Reading 79, Writing 80, Speaking 90	2024
GRE (General) Verbal Reasoning 140, Quantitative Reasoning 163 (out of 170)	2021
IELTS Academic Overall 7.5/9 (CEFR C1) - Listening 8.5, Reading 7.5, Writing 6.5, Speaking 7.0	2021

LICENSES & CERTIFICATES

Judo Coaching Certificate	2018
Judo Black Belt, 1st Dan	2015
Swimming Pool Lifeguard Certificate	2011
Table Tennis Coaching Certificate	2009

REFERENCES

[Giuk Lee](#)

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[Yong Ung Kwon](#)

Associate Professor, College of Sport Sciences, Chung-Ang University

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[Saeed Ilbeigi](#)

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